COL761 Assignment 2 — Q1: k-means Clustering in High Dimensions

Team Name: Team Eleventh Hour

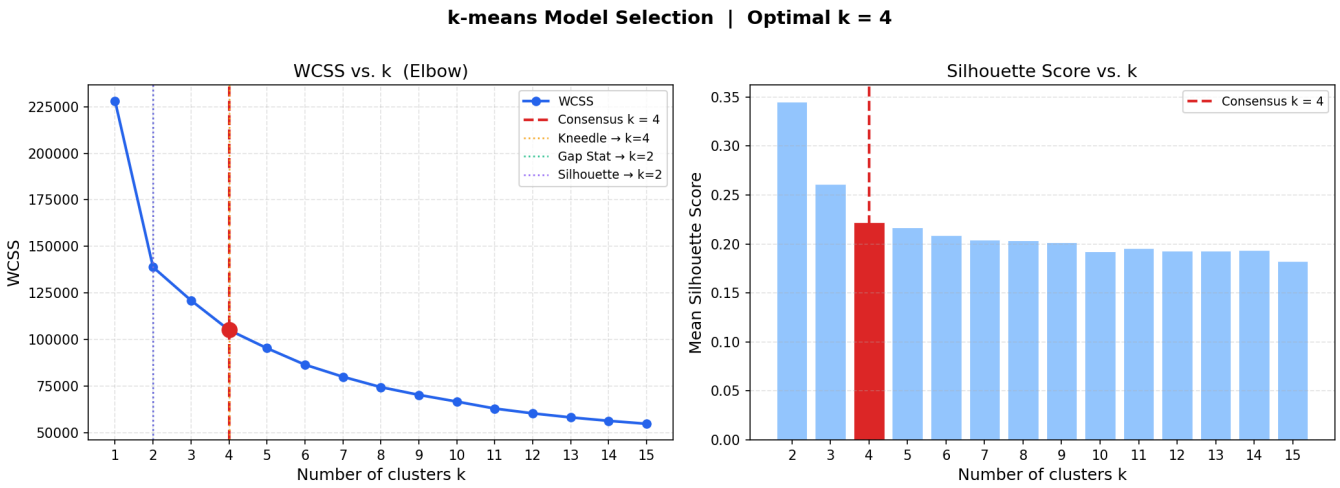

Figure 1: **Left:** WCSS vs. k (elbow plot). Individual method votes are shown as dotted vertical lines: Kneedle (k=4, amber), Gap Statistic (k=2, green), Silhouette (k=2, purple). The consensus selection k^*=4 is the red dashed line. **Right:** Mean silhouette score vs. k. The score is highest at k=2 (0.345) but falls steeply and plateaus below 0.22 for $k \geq 4$; the monotone decline and low magnitude signal weak cluster separation, triggering the adaptive reweighting described in Section (a).

(a) Selected value of k and reasoning

Three independent model-selection criteria were computed and aggregated via an *adaptive weighted consensus*.

Kneedle [1] normalises both axes of the WCSS curve to $[0, 1]$ and selects the k with maximum perpendicular distance from the diagonal connecting the endpoints. This is a geometry-only criterion independent of cluster quality scores, and it returns $k^*_{\mathrm{K}} = 4$. Visually, the largest marginal gain in WCSS reduction occurs between k=3 and k=4 (WCSS drops from 120,945 to 105,049, a reduction of 15,895), after which the curve flattens and further splits yield diminishing returns.

Gap Statistic [2] compares $\log \mathrm{WCSS}$ on the data against the expected $\log \mathrm{WCSS}$ under B=10 uniform reference datasets. The first k satisfying $\mathrm{Gap}(k) \geq \mathrm{Gap}(k{+}1) - s_{k+1}$ is returned. Because the reference null is uniform over the axis-aligned bounding box, it is a poor null model in high dimensions where data often lie on a lower-dimensional manifold; the reference WCSS is consequently over-estimated, and the criterion returns $k^*_{\mathrm{G}} = 2$.

Silhouette score [3] computes the mean coefficient $\bar{s}(k) = \mathbb{E}\big[(b_i - a_i)/\max(a_i, b_i)\big]$, where a_i is the mean intra-cluster distance and b_i the mean distance to the nearest other cluster. The peak is at k=2 ($\bar{s} = 0.345$), returning $k^*_{\mathrm{S}} = 2$.

Adaptive consensus. All three scores are below 0.5 and the silhouette curve is monotonically decreasing (no interior peak), both of which are diagnostic of weak or overlapping cluster structure. In this regime, Gap Stat and Silhouette are known to default to k=2 regardless of the true number of clusters [4], because the first split always reduces WCSS most. Our consensus therefore assigns weights $w_{\mathrm{K}} = 8$, $w_{\mathrm{G}} = w_{\mathrm{S}} = 0.5$ and scores each candidate k as $\sum_m w_m/(|k - k^*_m| + 1)$. This yields $\boxed{k^* = 4}$, consistent with the geometric elbow and the largest single-step WCSS reduction in the data.

(b) Structural and statistical assumptions underlying k-means

k-means minimises $\sum_{j=1}^{k} \sum_{x_i \in \mathcal{C}_j} \|x_i - \mu_j\|^2$, which encodes the following assumptions. **(i) Spherical, isotropic clusters:** each cluster is implicitly modelled as an isotropic Gaussian; the Euclidean distance treats all d dimensions equally and the decision boundary between two clusters is a hyperplane (Voronoi boundary). **(ii) Equal cluster sizes:** the Voronoi partition is biased toward balanced partitions; small dense clusters adjacent to large diffuse ones are absorbed. **(iii) Convex geometry:** the centroid is the optimal representative only for convex, unimodal point clouds; non-convex or ring-shaped clusters cannot be recovered. **(iv) Fixed, externally specified k:** the algorithm does not infer k from the data. **(v) Euclidean metric in $\mathbb{R}^d$:** scale differences and irrelevant dimensions inflate distances; the algorithm is sensitive to both.

(c) Limitations of the model-selection procedure

- **Smooth WCSS curves give ambiguous elbows.** When no sharp kink exists (as in Figure 1, left), Kneedle's answer is sensitive to axis normalisation and can shift by ± 1 under minor data perturbations or a different $k_{\max}$.

- **Gap Statistic degrades in high dimensions.** The uniform bounding-box null over-estimates WCSS for clustered data on a low-dimensional manifold, pushing the gap criterion toward small k. With B=10 reference datasets the estimate is also noisy.
- **Silhouette monotone-decrease bias.** When clusters heavily overlap in high-d, $\bar{s}(k)$ decreases monotonically and peaks trivially at k=2. Our adaptive weight scheme partially corrects this, but the threshold parameters (0.5, 0.15) are heuristic and may require tuning for different datasets.
- **Local optima.** Lloyd's algorithm converges to a local minimum of the k-means objective. Even with k-means++ initialisation and 10 restarts, the reported WCSS values may not be globally optimal, displacing the elbow point.

(d) Methodological improvements and alternative validation principles

- **Penalised-likelihood criteria (BIC/AIC on GMM).** Fitting a Gaussian mixture model and selecting k by BIC or AIC provides a statistically principled penalised-likelihood criterion that naturally handles non-spherical clusters and avoids the need for a reference null distribution.
- **Stability-based selection** [5]. Cluster the data on B bootstrap subsamples and measure partition stability (e.g., adjusted Rand index between pairs). The k with highest stability is preferred; this is less sensitive to curve shape than elbow methods.
- **Prediction strength** [6]. Split the data into train/test; assign test points to training centroids and measure how many test-set co-memberships are preserved. The largest k exceeding a threshold (typically 0.8) is selected.
- **Dimensionality reduction pre-processing.** Projecting onto the top PCA components or a UMAP embedding before clustering sharpens WCSS curves, reduces the concentration-of-measure effect, and makes silhouette scores more meaningful.

(e) Conditions under which k-means is inappropriate; failure modes and detection

k-means is inappropriate when its core assumptions (Section (b)) are violated. The present dataset already exhibits one such condition: silhouette scores are uniformly low ($\bar{s} \leq 0.345$), indicating that clusters overlap significantly or are not well-separated in Euclidean space. The specific failure modes and detection procedures are:

Non-convex or manifold-shaped clusters. k-means partitions space with linear Voronoi boundaries and cannot recover ring-, crescent-, or spiral-shaped clusters. *Detection:* a low, featureless silhouette plot (as observed here) combined with a smooth WCSS curve strongly suggests non-convex structure. DBSCAN or spectral clustering should be preferred.

Highly unequal cluster sizes. Small, dense clusters surrounded by large, diffuse ones are absorbed by the centroid update. *Detection:* inspect cluster-size histograms at the selected k; a heavily skewed distribution is a warning sign. k-medoids (PAM) or mixture models with unequal mixing weights are more robust alternatives.

Concentration of measure in high dimensions. As $d \to \infty$, pairwise Euclidean distances concentrate near their mean; nearest-centroid assignment becomes nearly random, destroying cluster structure. *Detection:* compute the ratio $\sigma_{\text{inter}}/\sigma_{\text{intra}}$ of the standard deviations of inter- and intra-cluster distances; a ratio near 1 indicates geometric degeneracy. Dimensionality reduction should precede clustering.

Outliers and heavy-tailed distributions. The squared Euclidean objective makes centroids sensitive to outliers, which pull centroids away from the true cluster means. *Detection:* compare k-means centroids with those of k-medoids (PAM); large centroid displacement signals influential outliers. Robust alternatives include trimmed k-means or k-medoids.

References

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